

CAN PRICE GET THE MONKEY OFF OUR BACK? A META-ANALYSIS OF ILLICIT DRUG DEMAND

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SUMMARY

Because of the increased availability of price data over the past 15 years, several studies have estimated the demand for illicit drugs, providing 462 estimates of the price elasticity. Results from estimating several meta-regressions reveal that these price elasticity estimates are influenced by a number of study characteristics. For instance, the price elasticity differs across drugs, with its absolute value being smallest for marijuana, compared with cocaine and heroin. Furthermore, price elasticity estimates are sensitive to whether demand is modeled in the short-run or the long-run, measures of quantity and price, whether or not alcohol and other illicit drugs are included in the specification of demand, and the location of demand. However, a number of other factors, including the functional form of demand, several specification issues, the type of data and method used to estimate demand, and the quality of the publication outlet, have less influence on the price elasticity. Copyright © 2013 John Wiley & Sons, Ltd.

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1. INTRODUCTION

In light of the economic cost of illicit drug use, recently estimated at nearly \$200 billion per year for the United States (National Drug Intelligence Center, 2011), there has been much debate regarding the efficacy of drug control policy. With the bulk of effort directed toward reducing supply (e.g., crop eradication, drug seizures, and penalties levied on drug sellers), any policy evaluation requires an understanding of the price elasticity of demand (see White and Luksetich, 1983; Grossman *et al.*, 2002; Becker *et al.*, 2004). Fortunately, the increased availability of price data over the past 15 years has led to numerous estimates of the price elasticity, most commonly for marijuana, cocaine, and heroin. However, because there is disagreement in the literature, with some studies estimating price elasticities that lie in the inelastic range (e.g., Van Ours, 1995; DeSimone and Farrelly, 2003; Dave, 2006), whereas others estimate price elasticities that lie in the elastic range (e.g., Petry and Bickel, 1998; Caulkins, 2001; Chalmers *et al.*, 2010), it remains a challenge to evaluate supply-side policies, for price changes resulting from supply shifts have less (more) influence on drug consumption when the absolute price elasticity of demand is lower (higher).

To date, no study has systematically reviewed the literature on the price elasticity of illicit drugs, and so this paper quantitatively surveys the literature by estimating several meta-regressions. Unlike traditional qualitative literature reviews, which can suffer from the subjective decision of the reviewer to discount certain studies or characteristics, by estimating a meta-regression model we rely on statistical tests to evaluate the importance of study characteristics in determining the price elasticity estimate. Questions addressed by our meta-analysis

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include the following: (i) Does the price elasticity differ across drugs?; (ii) Are short-run and long-run price elasticities different?; (iii) Do the measures of quantity and price influence the price elasticity?; (iv) Is the price elasticity sensitive to the chosen functional form of demand, as well as specification, data, and estimation issues?; And (v) does the quality of the publication outlet influence the reported price elasticity?

Briefly, across several meta-regressions, we find that the demand for marijuana is less responsive to price, compared with cocaine and heroin, and the price elasticity is higher in absolute value in the long-run than in the short-run. Other findings include the following: (i) the decision to use drugs (i.e., demand at the extensive margin) is more responsive to price than the decision of how much to use (i.e., demand at the intensive margin); (ii) the use of hypothetical price data common among behavioral economic studies is associated with much inflated price elasticities; (iii) the estimated price elasticity is sensitive to whether or not the specification of demand controls for the roles of substitutes and/or complements; and (iv) the price elasticity in the United States is smaller in absolute value. Nonetheless, other characteristics, including the functional form of demand, several specification issues, the type of data and method used to estimate demand, and the quality of the publication outlet, do not appreciably influence the price elasticity.

In the sections that follow, a summary of the literature is provided, which is followed by a discussion of the meta-regression model and estimation results. The paper concludes with a summary of the findings.

2. LITERATURE SUMMARY

Because the focus of this study is on the price responsiveness of illicit drug use, a search was undertaken to identify studies that provide estimates of the price elasticities of illicit drugs. Specifically, in addition to utilizing several online search engines (i.e., EconLit, Social Science Research Network, and Google Scholar), as well as examining the reference sections of all studies, we identified 42 studies that collectively provide 462 estimates of the price elasticities of marijuana, cocaine, and heroin (Table I).¹

Perusing Table I, with the exception of Nisbet and Vakil's (1972) study of marijuana use among students at the University of California (at Los Angeles), price elasticity estimates are from studies published since the mid-1990s. This is largely due to the absence of price data prior to the 1990s.² Although the median price elasticity estimate across all 462 estimates is -0.33 , implying on average that illicit drug demand is not very sensitive to price, as seen in Table I, there is substantial variation in the literature. For instance, the majority of studies of marijuana demand estimate absolute price elasticities that lie below the median, whereas studies of cocaine and heroin commonly estimate absolute price elasticities that lie above the median. Indeed, behavioral economic studies of cocaine and heroin (e.g., Petry and Bickel, 1998; Petry, 2001; Sumnall *et al.*, 2004; Jofre-Bonet and Petry, 2008; Chalmers *et al.*, 2009; Chalmers *et al.*, 2010) often obtain median price elasticities that lie in the elastic range. Also indicated in Table I, the vast majority of the 462 estimated price elasticities correspond to the US market.

Although Table I shows that studies differ according to the drug studied, as well as the location of demand, there are other notable differences in the literature. For instance, when it comes to quantity demanded, several studies

¹ A few studies were excluded because the reported price elasticities do not measure the impact of own-price on own-drug use. For instance, Silverman and Spruill's (1977) study was excluded because it provides indirect estimates of the price elasticity of heroin by analyzing crime patterns, not own-drug use. The study by Corman *et al.* (2005) was also excluded because it provides estimates of the impact of cocaine and heroin prices on overall illicit drug use, not own-drug use. Furthermore, although a few studies provide estimates of the price elasticities of methamphetamine (Sumnall *et al.*, 2004; Chalmers *et al.*, 2009; Chalmers *et al.*, 2010), ecstasy (Sumnall *et al.*, 2004), and valium (Petry and Bickel, 1998; Petry, 2001), because the number of estimates is small, in this study we focus on the price elasticities of marijuana, cocaine, and heroin, which are the three most commonly studied illicit drugs.

² There are several sources of illicit drug price data. First, a number of studies rely on price data collected from covert purchases of illicit drugs on behalf of law enforcement agencies. Second, some studies use price data collected from historical records pertaining to periods when illicit drug sales were legal but regulated. Third, other studies obtain price data by asking drug users how much they pay for drug purchases (i.e., self-reported price data). Finally, behavioral economic studies by-pass the problem of missing price data by asking individuals in an experimental setting how they would respond to hypothetical price changes.

Table I. Overview of studies

Study (year published)	Drug studied	Country	Number of observations (median price elasticity)
1. Bretteville-Jensen (1999)	H	Norway	6 (−0.96)
2. Bretteville-Jensen (2006)	H	Norway	2 (−0.55)
3. Bretteville-Jensen and Sutton (1996)	H	Norway	2 (−0.72)
4. Bretteville-Jensen and Bjørn (2003)	H	Norway	14 (−0.90)
5. Bretteville-Jensen and Bjørn (2004)	H	Norway	24 (−0.90)
6. Caulkins (1995)	C, H	US	12 (−0.34)
7. Caulkins (2001)	C, H	US	2 (−1.07)
8. Chalmers <i>et al.</i> (2009)	H	Australia	2 (−2.11)
9. Chalmers <i>et al.</i> (2010)	H	Australia	4 (−1.82)
10. Chaloupka <i>et al.</i> (1999)	C	US	16 (−0.42)
11. Chandra and Swoboda (2008)	M, H	India	4 (−0.52)
12. Clements and Daryal (2005)	M	Australia	1 (−0.69)
13. Crane <i>et al.</i> (1997)	C	US	2 (−0.46)
14. Dave (2006)	C, H	US	58 (−0.42)
15. Dave (2008)	C, H	US	54 (−0.52)
16. DeSimone (1998)	C	US	2 (−0.42)
17. DeSimone (2002)	C	US	2 (−1.26)
18. DeSimone (2005)	C	US	1 (−0.14)
19. DeSimone and Farrelly (2003)	M, C	US	28 (−0.10)
20. Grossman (2005)	M, C, H	US	13 (−0.27)
21. Grossman and Chaloupka (1998)	C	US	36 (−0.45)
22. Jofre–Bonet and Petry (2008)	C, H	US	8 (−0.90)
23. Liu <i>et al.</i> (1999)	H	Taiwan	14 (−0.27)
24. Nisbet and Vakil (1972)	M	US	4 (−0.30)
25. Pacula <i>et al.</i> (2001)	M	US	30 (−0.24)
26. Petry (2001)	C	US	1 (−1.15)
27. Petry and Bickel (1998)	H	US	5 (−1.06)
28. Ramful and Zhao (2009)	M	Australia	5 (−0.15)
29. Rhodes <i>et al.</i> (2000)	M, C, H	US	5 (−0.26)
30. Roddy and Greenwald (2009)	H	US	1 (−0.64)
31. Saffer and Chaloupka (1999a)	C, H	US	20 (−0.59)
32. Saffer and Chaloupka (1999b)	C, H	US	14 (−0.61)
33. Saffer and Dave (2005)	C	US	6 (−0.57)
34. Sumnall <i>et al.</i> (2004)	C	UK	1 (−2.44)
35. Van Luijk and Van Ours (2001)	H	Indonesia	4 (−0.22)
36. Van Ours (1995)	H	Indonesia	28 (−0.61)
37. Van Ours and Williams (2007)	M	Australia	4 (−0.60)
38. Williams (2004)	M	Australia	1 (−0.18)
39. Williams <i>et al.</i> (2004)	M	US	2 (−0.22)
40. Williams <i>et al.</i> (2006)	M, C	US	20 (−0.17)
41. Williams and Mahmoudi (2004)	M	Australia	3 (−0.44)
42. Zhao and Harris (2004)	M	Australia	1 (−0.21)
Total			462 (−0.33)

M, marijuana; C, cocaine; H, heroin. A few studies estimate the demand for opium (i.e., Van Ours, 1995; Liu *et al.*, 1999; Van Luijk and Van Ours, 2001; Chandra and Swoboda, 2008), which for the analysis is classified as heroin, because opium is processed into heroin.

(e.g., Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999a; DeSimone and Farrelly, 2003; Ramful and Zhao, 2009) rely on survey data to estimate the impact of price on the self-reported decision to use drugs (coded as 1 if the individual participates in drug use and 0 if the individual does not participate in drug use). As such, these studies focus on the extensive margin and therefore report ‘participation’ elasticities. Yet, other studies focus on the intensive margin by estimating the impact of price on the volume of drug use, thus treating consumption not as dichotomous but as continuous. Critics of studies that rely on self-reported drug use (e.g., Dave, 2006; Dave, 2008), however, argue that there may be inaccuracies if individuals are less inclined to report their true use of drugs. As an alternative, these studies objectively measure illicit drug consumption using a proxy, namely emergency room visits tied to a specific drug (e.g., Crane *et al.*, 1997; Caulkins, 2001; Grossman, 2005; Dave, 2006) or the percent of arrestees testing positive for a specific drug (e.g., Crane *et al.*, 1997; Rhodes *et al.*, 2000; Dave, 2008).

Further differences in the literature are tied to specific features of the illicit drug demand model. For instance, Chaloupka *et al.* (1999), Pacula *et al.* (2001), and DeSimone and Farrelly (2003) estimate the youth demand for illicit drugs, and in several specifications report price elasticities in absolute value which lie below the overall median.³ Furthermore, although some studies explore gender differences in illicit drug demand (e.g., Bretteville-Jensen, 1999; Saffer and Chaloupka, 1999b; Williams *et al.*, 2006), other studies (e.g., Van Ours, 1995; Grossman and Chaloupka, 1998; Liu *et al.*, 1999) address the addictive nature of illicit drugs by including either lagged consumption (i.e., myopic addiction) or both lagged and lead consumption (i.e., rational addiction) in the specification of demand. Also, a number of studies address relationships between drugs by controlling for the influence of alcohol and/or other illicit drugs on the demand for a particular drug. For instance, Saffer and Chaloupka (1999a) find illicit drugs are complements to each other, whereas Ramful and Zhao (2009) find they are substitutes.

3. META-REGRESSION MODEL

3.1. Specification

Similar to meta-analyses of the price elasticities of other goods, such as cigarettes (Gallet and List, 2003), water (Espey *et al.*, 1997; Dalhuisen *et al.*, 2003), gasoline (Espey, 1998), alcohol (Gallet, 2007; Wagenaar *et al.*, 2009), and meat (Gallet, 2010), we quantitatively explore the sensitivity of illicit drug price elasticities to a broad range of study characteristics. Specifically, because the typical study in Table I reports multiple price elasticities, we estimate a series of panel data meta-regressions, given by the following:

$$E_{ij} = \alpha + \beta X_{ij} + u_{ij}, \quad (1)$$

where E_{ij} denotes the j th price elasticity estimate taken from study i , α is an intercept common across i , X_{ij} accounts for study characteristics, β is a vector of coefficients, and u_{ij} is an error term.

In light of the discussion in the previous section, we include in X_{ij} dummy variables to control for commonly encountered study characteristics (i.e., each dummy variable is set equal to 1 if a particular study characteristic applies to the price elasticity estimate and 0 otherwise). First, we include dummy variables to control for the type of illicit drug consumed (i.e., marijuana, cocaine, or heroin). Second, we use dummy variables to account for differences in how quantity is measured (i.e., quantity measured in terms of participation; or quantity measured in a continuous manner; or quantity measured by the proxy of emergency room visits or arrestees testing positive for a specific drug), as well as how price is measured (i.e., price data obtained from covert purchases; price data obtained from historical records; self-reported price data obtained from drug users; or hypothetical price data used in behavioral economic studies). Third, we use dummy variables to control for a variety of specification issues, including (i) whether or not the price elasticity estimate comes from a demand specification that focuses on youths, males, or females, (ii) whether or not the demand specification controls for either myopic or rational addiction, and (iii) whether or not the demand specification accounts for the influence of alcohol and/or other illicit drugs on the demand for a particular drug.⁴

Similar to other meta-analyses of price elasticity (e.g., Espey, 1998; Dalhuisen *et al.*, 2003; Gallet and List, 2003; Gallet, 2010), we also include in X_{ij} dummy variables that control for a number of other demand

³DeSimone and Farrelly (2003) offer several arguments for why youths may be less responsive to price. It may be, for instance, that youths use money from their parents to purchase illicit drugs, or share drugs with other youths, and thus are less sensitive to price. Furthermore, because the share of marijuana among drug users is higher for youths (see Johnson and Gerstein, 1998), it may be that youths are less responsive to price because they predominantly consume a drug, which as suggested in Table I, has a lower absolute price elasticity.

⁴Failing to account for the role of alcohol and other illicit drugs in the specification of demand could increase the likelihood of omitted variable bias. That is, depending on whether alcohol and other illicit drugs are substitutes or complements to a particular drug, leaving one or both out of the demand equation could sway the price elasticity estimate for that drug one way or the other.

characteristics. Specifically, in addition to controlling for whether the elasticity pertains to the short-run or the long-run (i.e., whether the demand equation only includes contemporaneous variables or also includes non-contemporaneous variables), we control for the functional form of illicit drug demand (i.e., whether demand is linear, double-log, semi-log, or some other form). Also, we control for the type of data used to estimate demand (i.e., whether the data is cross-sectional, time-series, or panel), the location of demand (i.e., whether demand pertains to the United States or elsewhere), and the method used to estimate demand (i.e., ordinary least squares (OLS), maximum likelihood estimation, instrumental variables, or generalized least squares).

Finally, meta-analyses often control for the quality of the publication outlet (e.g., Dalhuisen *et al.*, 2003; Gallet and List, 2003; Gallet, 2010). Because papers published in top-ranked journals likely go through a more strenuous review process, which may influence the reported price elasticities, we control for whether or not the price elasticity appears in a top 36 economics journal (as defined by Scott and Mitias, 1996) or the top-ranked *Journal of Health Economics*. Variable names, definitions, and frequencies are provided in Table II.⁵

3.2. Estimation issues

There are several issues concerning the estimation of Eq. (1). First, because several categories of study characteristics encompass all observations, to avoid perfect multicollinearity, several of the variables in Table II are omitted from X_{ij} (i.e., marijuana, short-run, continuous, self-reported, other form, panel, and OLS). Along with setting all included dummy variables equal to zero, this defines the baseline upon which the results are compared.⁶ Second, as an aid to determine the functional form of the meta-regression, Eq. (1) was initially estimated using a Box-Cox transformation of the price elasticity. However, the estimated Box-Cox parameter was significantly different from 0 (i.e., semi-log form), but not 1 (i.e., linear form), and so the linear form was used throughout.

Third, there are several unusual price elasticity estimates. In particular, Caulkins (1995), Saffer and Chaloupka (1999a), Grossman (2005), and DeSimone and Farrelly (2003) report 10 price elasticity estimates that are positive. Such estimates are often excluded from meta-analyses of price elasticity as being unrealistic (see Espey *et al.*, 1997; Dalhuisen *et al.*, 2003; Gallet, 2010). Ten other price elasticity estimates are more than three standard deviations from the mean.⁷ Accordingly, to see whether or not the results are sensitive to potential outliers, we remove these 20 observations from the full sample and therefore also estimate Eq. (1) with a reduced sample of 442 observations.⁸

Fourth, we explore how sensitive the estimation of Eq. (1) is to different panel treatments. In particular, in addition to estimating Eq. (1) with no treatment for panel effects (i.e., OLS), we explore whether or not study characteristics beyond those contained in X_{ij} might influence the price elasticity by estimating a random effects (RE) version of Eq. (1).⁹ Furthermore, Table I indicates that seven lead authors (i.e., Bretteville-Jensen, Dave, DeSimone, Grossman, Saffer, Van Ours, and Williams) have published multiple studies accounting for over 70% of the estimated price elasticities. Similar to Gallet (2007), we also account for whether or not the price

⁵According to Table II, the greatest attention in the literature has been given to estimating the demand for heroin. Also, across the 42 studies, while the majority focus on the US market, only a few have estimated the youth demand for illicit drugs, addressed gender differences in demand, or appeared in the *Journal of Health Economics*.

⁶Because the typical price elasticity is negative, a positive (negative) coefficient in Eq. (1) implies the respective study characteristic that decreases (increases) the absolute value of the price elasticity (i.e., making illicit drug demand less (more) responsive to price), relative to the baseline.

⁷These ten estimates are reported in Grossman and Chaloupka (1998), Saffer and Chaloupka (1999a), Bretteville-Jensen (1999), DeSimone (2002), Sumnall *et al.* (2004), Grossman (2005), and Chalmers *et al.* (2009, 2010), with those furthest from the mean coming from the behavioral economic studies of Sumnall *et al.* (2004) and Chalmers *et al.* (2009, 2010).

⁸We also followed a procedure outlined by Bellavance *et al.* (2009) to identify outliers. Specifically, we split the price elasticities into quartiles and then constructed the following: (observed value—median)/(interquartile range). Bellavance *et al.* (2009) deem as outliers any observation for which this construction exceeds three in absolute value. In our case, the highest absolute value was 1.35, which suggests the presence of outliers is minor. Nonetheless, we do estimate Eq. (1) with the reduced sample of 442 observations as a robustness check.

⁹Several variables in X_{ij} are constant within each study (e.g., the variables controlling for publication characteristics), which prevents the use of fixed effects to estimate Eq. (1).

Table II. Summary of study characteristics

Variable	Definition	Share of observation	Share of studies
Illicit drug			
Marijuana	= 1 if price elasticity corresponds to marijuana	82/462	13/42
Cocaine	= 1 if price elasticity corresponds to cocaine	183/462	20/42
Heroin	= 1 if price elasticity corresponds to heroin	197/462	22/42
Time frame			
Short-run	= 1 if price elasticity corresponds to the short-run	391/462	42/42
Long-run	= 1 if price elasticity corresponds to the long-run	71/462	8/42
Quantity			
Participation	= 1 if price elasticity corresponds to decision to participate in drug use	170/462	18/42
Continuous	= 1 if price elasticity corresponds to how much drugs to consume	152/462	20/42
Proxy	= 1 if price elasticity corresponds to proxy for quantity	140/462	6/42
Price			
Covert	= 1 if price elasticity estimated with price data from covert purchases	332/462	29/42
Historical	= 1 if price elasticity estimated with price data from historical records	50/462	4/42
Hypothetical	= 1 if price elasticity estimated with hypothetical price data	23/462	6/42
Self-Reported	= 1 if price elasticity estimated with buyer self-reported price data	57/462	9/42
Functional form			
Linear	= 1 if price elasticity from linear demand form	78/462	7/42
Semi-log	= 1 if price elasticity from semi-log demand form	158/462	7/42
Double-log	= 1 if price elasticity from double-log demand form	73/462	14/42
Other form	= 1 if price elasticity from other form of demand	153/462	20/42
Specification			
Alcohol	= 1 if price elasticity from demand that includes alcohol	97/462	11/42
Other drug	= 1 if price elasticity from demand that includes other illicit drugs	203/462	20/42
Youth	= 1 if price elasticity from youth demand	62/462	4/42
Male	= 1 if price elasticity from male demand	11/462	5/42
Female	= 1 if price elasticity from female demand	8/462	3/42
Addiction	= 1 if price elasticity from demand that controls for addiction	82/462	6/42
Data			
Cross-sectional	= 1 if price elasticity estimated using cross-sectional data	195/462	27/42
Time series	= 1 if price elasticity estimated using time-series data	47/462	6/42
Panel	= 1 if price elasticity estimated using panel data	220/462	9/42
US	= 1 if price elasticity estimated using US data	342/462	24/42
Estimation			
OLS	= 1 if price elasticity estimated using ordinary least squares	180/462	20/42
MLE	= 1 if price elasticity estimated using maximum likelihood	161/462	22/42
IV	= 1 if price elasticity estimated using instrumental variables	106/462	9/42
GLS	= 1 if price elasticity estimated using generalized least squares	15/462	4/42
Publication			
Top 36 journal	= 1 if price elasticity published in a top 36 economics journal	146/462	8/42
J. Health Econ.	= 1 if price elasticity published in the <i>Journal of Health Economics</i>	112/462	4/42

elasticities are author-sensitive.¹⁰ To do so, we estimate OLS and RE regressions under two further scenarios, one that includes author controls (i.e., includes seven additional dummy variables accounting for these respective authors, with each dummy variable set equal to 1 if the price elasticity estimate is from a study published by the associated author, 0 if not) and one that excludes these author controls.

4. ESTIMATION RESULTS

4.1. Meta-regression results

The full and reduced sample results from estimating Eq. (1) are provided in Tables III and IV, respectively. Beginning with the full sample results, the Lagrange multiplier test statistic reported at the bottom

¹⁰If particular authors use unique data sets or model specifications, then the price elasticity estimate may be unique to the author.

Table III. Full sample meta-regression results

Variables	Author controls excluded		Author controls included	
	OLS	RE	OLS	RE
Illicit drug				
Cocaine	−0.41 (6.69)***	−0.28 (4.61)***	−0.32 (4.30)***	−0.22 (3.26)***
Heroin	−0.39 (6.30)***	−0.22 (3.83)***	−0.27 (3.36)***	−0.16 (2.16)**
Time frame				
Long-run	−0.23 (6.76)***	−0.23 (7.42)***	−0.24 (7.46)***	−0.24 (7.66)***
Quantity				
Participation	−0.19 (1.86)*	−0.35 (3.45)***	−0.21 (1.78)*	−0.40 (3.22)***
Proxy	0.15 (1.60)	−0.12 (0.92)	−0.03 (0.29)	−0.26 (1.31)
Price				
Covert	−0.02 (0.33)	0.23 (2.12)**	−0.08 (0.61)	0.25 (1.06)
Historical	0.41 (3.72)***	0.46 (2.76)***	0.29 (0.97)	0.55 (0.98)
Hypothetical	−0.77 (4.73)***	−0.68 (2.90)***	−0.94 (9.57)***	−0.63 (3.77)***
Functional form				
Linear	−0.18 (1.13)	−0.20 (0.76)	−0.15 (0.71)	−0.22 (0.95)
Semi-log	0.09 (0.71)	0.03 (0.19)	0.01 (0.09)	−0.03 (0.24)
Double-log	−0.14 (0.89)	−0.10 (0.41)	−0.07 (0.46)	−0.12 (0.58)
Specification				
Alcohol	−0.13 (2.99)***	−0.10 (1.85)*	−0.11 (1.80)*	−0.09 (1.43)
Other drug	0.12 (3.32)***	0.11 (2.24)**	0.09 (1.99)**	0.11 (1.94)*
Youth	0.05 (0.66)	0.14 (1.81)*	0.08 (1.22)	0.18 (2.61)***
Male	−0.20 (1.63)	−0.18 (1.15)	−0.25 (1.84)*	−0.18 (1.05)
Female	−0.07 (0.39)	−0.09 (0.45)	−0.14 (0.85)	−0.09 (0.50)
Addiction	0.10 (1.30)	−0.06 (0.54)	0.02 (0.25)	−0.11 (1.06)
Data				
Cross-sectional	0.03 (0.33)	0.01 (0.02)	0.04 (0.22)	0.04 (0.22)
Time series	0.08 (1.02)	0.20 (1.56)	0.11 (1.10)	0.30 (1.94)*
US	0.54 (6.24)***	0.44 (4.07)***	0.47 (6.52)***	0.44 (4.00)***
Estimation				
MLE	0.10 (1.43)	0.21 (3.05)***	0.11 (1.34)	0.21 (1.48)
IV	0.01 (0.16)	0.04 (0.87)	0.05 (1.32)	0.05 (1.09)
GLS	0.10 (0.52)	0.08 (0.46)	0.16 (0.91)	0.09 (0.53)
Publication				
Top 36 journal	0.06 (0.86)	0.12 (0.72)	0.12 (0.76)	0.19 (0.58)
J. Health Econ.	0.10 (1.47)	0.19 (1.26)	0.10 (0.75)	0.19 (0.69)
# observations	462	462	462	462
Adj. R ²	0.46		0.48	
LM (<i>p</i> -value)		0.03 (0.87)		1.31 (0.25)
<i>F</i> (author controls)			6.13	2.48
(<i>p</i> -value)			(0.00)	(0.02)

MLE, maximum likelihood estimation; IV, instrumental variables; GLS, generalized least squares; RE, random effects; LM, Lagrange multiplier; OLS, ordinary least squares.

White (1980) standard errors are used to construct absolute *t*-statistics, which are provided in parentheses next to coefficient estimates. The results in the last two columns correspond to regressions that also include seven dummy variables to control for those authors who account for over 70% of the estimated price elasticities. *F* (author controls) corresponds to a test of the joint significance of the coefficients of these author control variables.

*Significant at 10% level; **Significant at 5% level; ***Significant at 1% level.

of Table III favors OLS over RE estimation, regardless of whether the author controls are included or excluded. As for the author controls, the coefficients of the seven author dummy variables are jointly significant in the OLS and RE regressions, thus suggesting there are additional features tied to these authors that influence the reported price elasticities.

The results are similar in many respects across all four meta-regressions, and so, rather than discuss each meta-regression separately, we focus on the general pattern of the results. To begin, the coefficients of the cocaine and heroin dummy variables in Table III are significantly negative in all meta-regressions,

Table IV. Reduced sample meta-regression results

Variables	Author controls excluded		Author controls included	
	OLS	RE	OLS	RE
Illicit drug				
Cocaine	−0.35 (5.74)***	−0.26 (4.48)***	−0.26 (3.77)***	−0.22 (3.42)***
Heroin	−0.35 (5.75)***	−0.24 (4.36)***	−0.23 (3.31)***	−0.19 (3.05)***
Time frame				
Long-Run	−0.21 (5.84)***	−0.22 (6.28)***	−0.22 (6.41)***	−0.22 (6.53)***
Quantity				
Participation	−0.19 (2.43)**	−0.29 (3.75)***	−0.21 (2.74)***	−0.30 (3.75)***
Proxy	0.17 (2.67)***	−0.04 (0.42)	−0.09 (1.02)	−0.18 (1.68)*
Price				
Covert	0.14 (2.00)**	0.30 (2.80)***	0.11 (1.12)	0.21 (1.56)
Historical	0.37 (4.74)***	0.37 (3.78)***	0.22 (1.20)	0.27 (1.11)
Hypothetical	−0.32 (3.29)***	−0.32 (2.17)**	−0.48 (3.81)***	−0.43 (2.30)**
Functional form				
Linear	−0.04 (0.35)	−0.08 (0.56)	0.01 (0.04)	−0.06 (0.37)
Semi-log	0.08 (1.17)	0.04 (0.38)	0.02 (0.28)	0.03 (0.37)
Double-log	−0.18 (2.23)**	−0.11 (0.90)	−0.05 (0.55)	−0.04 (0.29)
Specification				
Alcohol	−0.14 (3.62)***	−0.13 (2.72)***	−0.12 (3.00)***	−0.12 (2.56)**
Other Drug	0.06 (1.62)	0.08 (1.87)*	0.06 (1.74)*	0.08 (1.82)*
Youth	0.02 (0.45)	0.04 (0.60)	0.05 (0.95)	0.07 (1.18)
Male	−0.10 (0.55)	−0.17 (0.81)	−0.15 (0.80)	−0.18 (0.90)
Female	0.07 (0.57)	0.02 (0.14)	0.03 (0.28)	0.01 (0.11)
Addiction	0.07 (0.82)	−0.04 (0.42)	−0.01 (0.07)	−0.07 (0.67)
Data				
Cross-sectional	−0.01 (0.20)	0.01 (0.11)	0.01 (0.03)	0.03 (0.39)
Time series	−0.03 (0.28)	0.09 (0.76)	0.03 (0.32)	0.11 (1.02)
US	0.26 (4.14)***	0.19 (2.03)**	0.10 (0.97)	0.08 (0.63)
Estimation				
MLE	0.15 (1.64)	0.18 (1.61)	0.17 (2.19)**	0.19 (1.93)*
IV	0.01 (0.32)	0.04 (0.96)	0.07 (1.86)*	0.06 (1.54)
GLS	−0.04 (0.38)	0.01 (0.09)	0.03 (0.26)	0.04 (0.27)
Publication				
Top 36 journal	0.03 (0.76)	0.10 (1.08)	0.05 (0.50)	0.12 (0.73)
J. Health Econ.	0.03 (0.78)	0.10 (0.94)	−0.02 (0.18)	0.06 (0.36)
# Observations	442	442	442	442
Adj. R ²	0.44		0.47	
LM (<i>p</i> -value)		1.00 (0.32)		2.48 (0.12)
<i>F</i> (author controls)			7.29	7.09
(<i>p</i> -value)			(0.00)	(0.00)

MLE, maximum likelihood estimation; IV, instrumental variables; GLS, generalized least squares; RE, random effects; LM, Lagrange multiplier; OLS, ordinary least squares.

White (1980) standard errors are used to construct absolute *t*-statistics, which are provided in parentheses next to coefficient estimates. The results in the last two columns correspond to regressions that also include seven dummy variables to control for those authors who account for over 70% of the estimated price elasticities. *F* (author controls) corresponds to a test of the joint significance of the coefficients of these author control variables; *Significant at 10% level; **Significant at 5% level; ***Significant at 1% level.

which implies the baseline drug (marijuana) is least responsive to price.¹¹ Because marijuana is considered a soft drug, whereas cocaine and heroin are considered hard drugs, such results run counter to the expectation that hard drugs are more addictive and therefore less responsive to price. Although the literature is silent on why marijuana use is less responsive to price, there are several possible scenarios that could support such a result. First, according to Wickelgren (1997), it may be that marijuana is more

¹¹The coefficients of the cocaine and heroin dummy variables being similar in magnitude also imply that the corresponding price elasticities are similar, *ceteris paribus*. Interestingly, some studies of multiple drugs also find that marijuana is less responsive to price compared with cocaine and heroin (DeSimone and Farrelly, 2003; Williams *et al.*, 2006; Chandra and Swoboda, 2008).

addictive than its designation as a soft drug suggests, thus reducing its absolute price elasticity. Second, if the budget share of marijuana is smaller than that of cocaine and heroin, this will reduce the absolute price elasticity of marijuana.¹² Third, if there is a gateway effect, such that present soft drug use increases the likelihood of future hard drug use, then the typical hard drug user will have more experience in the illicit drug market. As such, they could have greater information on the location and availability of substitutes, thus increasing their price responsiveness.¹³ Fourth, because heroin and cocaine users are more often polydrug users, they may perceive greater substitutability among hard drugs, thus leading them to be more responsive to price.

Turning to the other characteristics of demand, as expected, the long-run illicit drug price elasticity is higher in absolute value compared with its short-run counterpart. We also find the following: (i) the measure of quantity demanded affects the price elasticity, as price has a greater impact on demand at the extensive margin (i.e., the decision to participate in drug use) compared with the intensive margin (i.e., the decision of how much to consume); (ii) the use of hypothetical price data, which is common in behavioral economic studies, is associated with a higher absolute price elasticity; (iii) accounting for the impact of alcohol (other illicit drugs) on own-drug demand inflates (deflates) the absolute price elasticity, which is consistent with the presence of omitted variable bias in studies that fail to control for the roles of substitutes and/or complements; and (iv) the demand for illicit drugs in the United States is less responsive to price compared with other countries.

For the remaining variables, though, the coefficients are not significantly different from zero across a majority of the meta-regressions, and so the associated characteristics have less influence on the price elasticity. Indeed, although we find when author controls are excluded that historical price data deflates the absolute price elasticity, and that youths are less responsive to price in the RE regressions, the remaining coefficients are insignificantly different from zero in most cases. Accordingly, the price elasticity is largely insensitive to such factors as the functional form of demand, whether the demand specification addresses gender differences or addiction, the nature of the data (be it cross-sectional, time-series, or panel), the method used to estimate demand, and the quality of the publication outlet.

There are few changes in the results when the outliers are removed from the full sample. For instance, as indicated at the bottom of Table IV, Lagrange multiplier tests continue to favor OLS over RE estimation, and the *F*-tests continue to support the inclusion of author controls in the meta-regressions. Also, we continue to find the following: (i) cocaine and heroin are more responsive to price; (ii) the absolute price elasticity is higher in the long-run; (iii) measuring quantity in terms of participation and eliciting consumer response to hypothetical changes in price inflates the absolute price elasticity; (iv) demand is more (less) responsive to price when alcohol (other illicit drugs) is included in the specification; and (v) the price elasticity in the United States is lower in absolute value. Furthermore, similar to the full sample results, we continue to find the price elasticity that is largely insensitive to the functional form of demand, whether the study addresses gender differences or controls for addiction in the specification of demand, the nature of the data used to estimate demand, the estimation method, and the quality of the publication outlet.¹⁴

4.2. Predicted price elasticities

To have a better sense of how study characteristics influence the illicit drug price elasticity, we use the meta-regression results to predict the price elasticities associated with those variables that have coefficients most often significantly different from zero in Tables III and IV. Specifically, for each of these respective study

¹²Although we lack budget share data, according to the Office of National Drug Control Policy (2001), throughout much of the 1980s and 1990s, the total expenditure on marijuana in the United States was below that of cocaine and marijuana. Indeed, total expenditure on cocaine was nearly fourfold higher than marijuana.

¹³See DeSimone (1998) and Van Ours (2003) for analyses of the gateway effect.

¹⁴However, there are some differences between the full sample and reduced sample results. For instance, in addition to differences in significance of certain coefficients, in all of the reduced sample meta-regressions, we now find the price elasticity for youths is insignificantly different from that of the general population.

characteristics, the corresponding predicted price elasticity is constructed by holding all study characteristic dummy variables equal to their means, with the exception of the dummy variables in each associated category, which are either set to zero or one. For example, the predicted price elasticity of the baseline drug marijuana is determined by setting all study characteristic dummy variables equal to their means, with the exception of the dummy variables corresponding to cocaine and heroin, both of which are set to zero. As for cocaine, its predicted price elasticity is determined by setting all study characteristic dummy variables equal to their means, except in this case the dummy variable for cocaine is set to one and the dummy variable for heroin is set to zero. Similar procedures are followed to construct other predicted price elasticities, which are provided in Table V.

A perusal of the price elasticities in Table V reveals several interesting results. First, although the predicted price elasticities of cocaine and heroin are similar in magnitude, indeed lying in the neighborhood of the price elasticities of cigarettes and alcohol (Gallet and List, 2003; Gallet, 2007; Wagenaar *et al.*, 2009), in the majority of cases, the predicted price elasticity of marijuana is nearly half that of cocaine and heroin.¹⁵ Second, short-run price elasticities, as well as price elasticities from specifications that include other illicit drugs in demand, lie below 0.50 in absolute value, whereas long-run price elasticities, coupled with price elasticities from specifications that include alcohol in demand, lie above 0.50 in absolute value. Third, compared with other predicted price elasticities, the absolute price elasticity is noticeably higher when quantity is measured in terms of participation, and price data is hypothetical.¹⁶ Fourth, the price elasticity corresponding to the United States is lower in magnitude compared with others in Table V.

5. CONCLUSION

Our quantitative survey indicates several characteristics of the literature that influence the price elasticity of illicit drugs. For instance, we not only find demand at the extensive margin is more responsive to price than at the intensive margin but also we find the demand for marijuana is much less responsive to price, compared with that of cocaine and heroin. Furthermore, behavioral economic studies that rely on hypothetical price data tend to report price elasticities that are higher in absolute value, whereas studies of the US market tend to report price elasticities that are lower in absolute value.

The results of this meta-analysis are useful in several ways. First, a quantitative survey of the literature on illicit drug demand is not only useful as a means of providing summary estimates of the price elasticity but also finding that there are key tendencies in the literature that sway price elasticities one way or the other suggests avenues for future research. For instance, studies may wish to examine why marijuana demand is less responsive to price, or why the decision to participate in drug use is more responsive to price, or why price elasticities reported in behavioral economic studies often lie in the elastic range. Second, by finding that a number of characteristics, such as the functional form of demand, several specification issues, the type of data and method used

¹⁵There is also little variability in the predicted price elasticities of cocaine and heroin, as they all fall in the range of -0.47 to -0.56 , whereas the predicted price elasticity of marijuana based on the OLS results is always lower in absolute value compared with the RE results, regardless of whether the full or reduced samples are used and whether author controls are included or excluded. Thus, for marijuana, the results are more sensitive to the method used to estimate the meta-regression model.

¹⁶Indeed, although the predicted price elasticity in most cases lies in the inelastic range, the use of hypothetical price data is associated with elasticities that typically lie in the elastic range. As mentioned previously, though, behavioral economic studies rely on hypothetical price data and also report price elasticities in the elastic range. Thus, our findings are consistent with the results reported in these behavioral economic studies. Interestingly, the predicted price elasticity associated with the use of hypothetical price data is lower in magnitude in the reduced sample, compared with the full sample. The reason for this is tied to the removal of several unusually large price elasticity estimates reported in the behavioral economic studies of Sumnall *et al.* (2004) and Chalmers *et al.* (2009, 2010). Future research may wish to consider why behavioral economic studies often estimate price elasticities far beyond those reported in other studies. Perhaps, for example, because these studies often rely on polydrug drug users as experiment subjects, these users are sufficiently educated about the drug market to be more sensitive to price. Alternatively, similar to nonmarket valuation experiments, perhaps there exists some degree of bias in an experiment subject's response to a hypothetical price change (e.g., see List and Gallet, 2001).

Table V. Predicted price elasticities

Variables	Author controls excluded		Author controls included	
	OLS	RE	OLS	RE
Full sample				
Illicit drug				
Marijuana	−0.15	−0.28	−0.23	−0.31
Cocaine	−0.56	−0.55	−0.55	−0.53
Heroin	−0.54	−0.50	−0.50	−0.47
Time frame				
Short-run	−0.44	−0.45	−0.44	−0.43
Long-run	−0.67	−0.68	−0.68	−0.67
Quantity				
Participation	−0.64	−0.67	−0.60	−0.64
Price				
Hypothetical	−1.24	−1.34	−1.35	−1.30
Specification				
Alcohol	−0.58	−0.57	−0.56	−0.54
Other Drug	−0.41	−0.42	−0.42	−0.41
Data				
US	−0.34	−0.37	−0.36	−0.38
Reduced sample				
Illicit drug				
Marijuana	−0.16	−0.26	−0.25	−0.29
Cocaine	−0.51	−0.52	−0.51	−0.51
Heroin	−0.51	−0.50	−0.48	−0.48
Time frame				
Short-run	−0.42	−0.43	−0.42	−0.42
Long-run	−0.63	−0.65	−0.64	−0.64
Quantity				
Participation	−0.63	−0.64	−0.56	−0.59
Price				
Hypothetical	−0.90	−1.03	−1.01	−1.04
Specification				
Alcohol	−0.56	−0.57	−0.54	−0.54
Other Drug	−0.41	−0.42	−0.42	−0.41
Data				
US	−0.38	−0.41	−0.42	−0.43

RE, random effects; OLS, ordinary least squares.

For the various study characteristics, predicted price elasticities are constructed by holding all study characteristic dummy variables equal to their respective means, with the exception of the category variables in question, whose respective dummy variables are set equal to either one or zero. For instance, the predicted price elasticity of heroin is obtained by setting all other dummy variables equal to their means, with the exception of the dummy variable for heroin (set equal to one) and the dummy variable for cocaine (set equal to zero).

to estimate demand, and the quality of publication outlet, have less influence on the price elasticity estimate, less concern needs to be given to these characteristics when selecting a price elasticity estimate from the literature.

Lastly, there are policy implications associated with our results. For instance, because the price elasticity is smallest in absolute value for marijuana, advocates of legalizing marijuana could argue that the increase in supply resulting from decriminalizing the sale of marijuana would lead to a modest increase in consumption (absent demand shifts), compared with cocaine and heroin. Thus, the benefits of legalizing marijuana might outweigh the costs. Also, given we find disparities in price elasticity estimates, this suggests a ‘one size fits all’ approach may be inappropriate when designing drug control policy. For instance, by finding participation price elasticities that are larger in absolute value, policies that increase price have a greater impact on an individual’s choice of whether or not to consume drugs, as opposed to how much to consume. Thus, policies that increase price may be most effective at discouraging initiation into drug use.

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REFERENCES

- Becker GS, Murphy KM, Grossman M. 2004. The economic theory of illegal goods: the case of drugs. NBER Working Paper 10976.
- Bellavance F, Dionne G, Lebeau M. 2009. The value of a statistical life: a meta-analysis with a mixed effects regression model. *Journal of Health Economics* **28**: 444–464.
- Bretteville-Jensen AL. 1999. Gender, heroin consumption and economic behaviour. *Health Economics* **8**: 379–389.
- Bretteville-Jensen AL. 2006. Drug demand – initiation, continuation and quitting. *De Economist* **154**: 491–516.
- Bretteville-Jensen AL, Biørn E. 2003. Heroin consumption, prices and addiction: evidence from self-reported panel data. *The Scandinavian Journal of Economics* **105**: 661–679.
- Bretteville-Jensen AL, Biørn E. 2004. Do prices count? A micro-econometric study of illicit drug consumption based on self-reported data. *Empirical Economics* **29**: 673–695.
- Bretteville-Jensen AL, Sutton M. 1996. Under the influence of the market: an applied study of illicitly selling and consuming heroin. Working paper.
- Caulkins JP. 1995. Estimating elasticities of demand for cocaine and heroin with data from the Drug Use Forecasting system. Working paper.
- Caulkins JP. 2001. Drug prices and emergency room mentions for cocaine and heroin. *American Journal of Public Health* **91**: 1446–1448.
- Chalmers J, Bradford D, Jones C. 2009. How do methamphetamine users respond to changes in methamphetamine price? *Crime and Justice Bulletin* **134**: 1–16.
- Chalmers J, Bradford D, Jones C. 2010. The effect of methamphetamine and heroin price on polydrug use: a behavioural economics analysis in Sydney, Australia. *The International Journal on Drug Policy* **21**: 381–389.
- Chaloupka FJ, Grossman M, Tauras JA. 1999. The demand for cocaine and marijuana by youth. In *The Economic Analysis of Substance Use and Abuse: An Integration of Econometrics and Behavioral Economic Research*, Chaloupka FJ, Grossman M, Bickel WK, Saffer H (eds.). National Bureau of Economic Research: Cambridge.
- Chandra S, Swoboda A. 2008. Are spatial variables important? The case of markets for multiple drugs in British Bengal. In *Geography and Drug Addiction*, Thomas YF, Richardson D, Cheung I (eds.). Springer: Berlin.
- Clements KW, Daryal M. 2005. The economics of marijuana consumption. In *The Demand for Alcohol, Tobacco, and Marijuana: International Evidence*, Selvanathan S, Selvanathan EA (eds.). Ashgate: England.
- Corman H, Noonan K, Reichman NE, Dave D. 2005. Demand for illicit drugs among pregnant women. In *Advances in Health Economics and Health Services Research*, Lindgren B, Grossman M (eds.), vol. 16. Elsevier: Amsterdam.
- Crane BD, Rivolo AR, Comfort GC. 1997. An empirical examination of counterdrug interdiction program effectiveness. Institute for Defense Analysis Working Paper P-3219.
- Dalhuisen JM, Florax RJGM, de Groot HLF, Nijkamp P. 2003. Price and income elasticities of residential water demand: a meta-analysis. *Land Economics* **79**: 292–308.
- Dave D. 2006. The effects of cocaine and heroin price on drug-related emergency department visits. *Journal of Health Economics* **25**: 311–333.
- Dave D. 2008. Illicit drug use among arrestees, prices and policy. *Journal of Urban Economics* **63**: 694–714.
- DeSimone J. 1998. Is marijuana a gateway drug? *Eastern Economic Journal* **24**: 149–164.
- DeSimone J. 2002. Illegal drug use and employment. *Journal of Labor Economics* **20**: 952–977.
- DeSimone J. 2005. Needle exchange programs and drug injection behavior. *Journal of Policy Analysis and Management* **24**: 559–577.
- DeSimone J, Farrelly MC. 2003. Price and enforcement effects of cocaine and marijuana demand. *Economic Inquiry* **41**: 98–115.
- Espey M. 1998. Gasoline demand revisited: an international meta-analysis of elasticities. *Energy Economics* **20**: 273–295.
- Espey M, Espey J, Shaw WD. 1997. Price elasticity of residential demand for water: a meta-analysis. *Water Resources Research* **33**: 1369–1374.

- Gallet CA. 2007. The demand for alcohol: a meta-analysis of elasticities. *The Australian Journal of Agricultural and Resource Economics* **51**: 121–135.
- Gallet CA. 2010. Meat meets meta: a quantitative review of the price elasticity of meat. *American Journal of Agricultural Economics* **92**: 258–272.
- Gallet CA, List JA. 2003. Cigarette demand: a meta-analysis of elasticities. *Health Economics* **12**: 821–835.
- Grossman M. 2005. Individual behaviours and substance use: the role of price. In *Advances in Health Economics and Health Services Research*, Lindgren B, Grossman M (eds.), vol. 16. Elsevier: Amsterdam.
- Grossman M, Chaloupka FJ. 1998. The demand for cocaine by young adults: a rational addiction approach. *Journal of Health Economics* **17**: 427–474.
- Grossman M, Chaloupka FJ, Shim K. 2002. Illegal drug use and public policy. *Health Affairs* **21**: 134–145.
- Jofre-Bonet M, Petry NM. 2008. Trading apples and oranges? Results of an experiment on the effects of heroin and cocaine price changes on addicts' polydrug use. *Journal of Economic Behavior and Organization* **66**: 281–311.
- Johnson RA, Gerstein DR. 1998. Initiation of use of alcohol, cigarettes, marijuana, cocaine, and other substances in U.S. birth cohorts since 1919. *American Journal of Public Health* **88**: 27–33.
- List JA, Gallet CA. 2001. What experimental protocol influence disparities between actual and hypothetical stated values? Evidence from a meta-analysis. *Environmental and Resource Economics* **20**: 241–254.
- Liu J, Liu J, Hammitt JK, Chou S. 1999. The price elasticity of opium in Taiwan, 1914–1942. *Journal of Health Economics* **18**: 795–810.
- National Drug Intelligence Center. 2011. *The Economic Impact of Illicit Drug Use on American Society*. Washington, DC.
- Nisbet C, Vakil F. 1972. Some estimates of price and expenditure elasticities of demand for marijuana among U.C.L.A. students. *The Review of Economics and Statistics* **54**: 473–475.
- Office of National Drug Control Policy. 2001. *What America's Users Spend on Illegal Drugs, 1988–2000*. Washington, DC.
- Pacula RL, Grossman M, Chaloupka FJ, O'Malley PM, Johnston LD, Farrelly MC. 2001. Marijuana and youth. In *Risky Behavior among Youths: An Economic Analysis*, Gruber J (ed.). University of Chicago Press: Chicago.
- Petry NM. 2001. A behavioral economic analysis of polydrug abuse in alcoholics: asymmetrical substitution of alcohol and cocaine. *Drug and Alcohol Dependence* **62**: 31–39.
- Petry NM, Bickel WK. 1998. Polydrug abuse in heroin addicts: a behavioral economic analysis. *Addiction* **93**: 321–335.
- Ramful P, Zhao X. 2009. Participation in marijuana, cocaine and heroin consumption in Australia: a multivariate probit approach. *Applied Economics* **41**: 481–496.
- Rhodes W, Johnston P, Han S, McMullen Q, Hozik L. 2000. *Illicit drugs: price elasticity of demand and supply*. Abt Associates Inc.: Cambridge, MA.
- Roddy J, Greenwald M. 2009. An economic analysis of income and expenditures by heroin-using research volunteers. *Substance Use & Misuse* **44**: 1503–1518.
- Saffer H, Chaloupka FJ. 1999a. The demand for illicit drugs. *Economic Inquiry* **37**: 401–411.
- Saffer H, Chaloupka FJ. 1999b. Demographic differentials in the demand for alcohol and illicit drugs. In *The Economic Analysis of Substance Use and Abuse: An Integration of Econometrics and Behavioral Economic Research*, Chaloupka FJ, Grossman M, Bickel WK, Saffer H (eds.). National Bureau of Economic Research: Cambridge.
- Saffer H, Dave D. 2005. Mental illness and the demand for alcohol, cocaine, and cigarettes. *Economic Inquiry* **43**: 229–246.
- Scott L, Mitias P. 1996. Trends in rankings of economics departments in the U.S.: an update. *Economic Inquiry* **34**: 378–400.
- Silverman LP, Spruill NL. 1977. Urban crime and the price of heroin. *Journal of Urban Economics* **4**: 80–103.
- Sumnall HR, Tyler E, Wagstaff GF, Cole JC. 2004. A behavioural economic analysis of alcohol, amphetamine, cocaine and ecstasy purchases by polysubstance misusers. *Drug and Alcohol Dependence* **76**: 93–99.
- Van Luijk EW, Van Ours JC. 2001. The effects of government policy on drug use: Java, 1875–1904. *Journal of Economic History* **61**: 1–18.
- Van Ours JC. 1995. The price elasticity of hard drugs: the case of opium in the Dutch East Indies, 1923–1938. *Journal of Political Economy* **103**: 261–279.
- Van Ours JC. 2003. Is cannabis a stepping-stone for cocaine? *Journal of Health Economics* **22**: 539–554.
- Van Ours JC, Williams J. 2007. Cannabis prices and dynamics of cannabis use. *Journal of Health Economics* **26**: 578–596.
- Wagenaar AC, Salois MJ, Komro KA. 2009. Effects of beverage alcohol price and tax levels on drinking: a meta-analysis of 1003 estimates from 112 studies. *Addiction* **104**: 179–190.
- White H. 1980. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica* **48**: 817–838.
- White MD, Luksetich WA. 1983. Heroin: price elasticity and enforcement strategies. *Economic Inquiry* **21**: 557–564.
- Wickelgren I. 1997. Marijuana: harder than thought? *Science* **276**: 1967–1968.
- Williams J. 2004. The effects of price and policy on marijuana use: what can be learned from the Australian experience? *Health Economics* **13**: 123–137.
- Williams J, Mahmoudi P. 2004. Economic relationship between alcohol and cannabis revisited. *The Economic Record* **80**: 36–48.

- Williams J, Pacula RL, Chaloupka FJ, Wechsler H. 2004. Alcohol and marijuana use among college students: economic complements or substitutes? *Health Economics* **13**: 825–843.
- Williams J, Pacula RL, Chaloupka FJ, Wechsler H. 2006. College students' use of cocaine. *Substance Use & Misuse* **41**: 489–509.
- Zhao X, Harris MN. 2004. Demand for marijuana, alcohol and tobacco: participation, levels of consumption and cross-section correlations. *The Economic Record* **80**: 394–410.